

GPIO assignment

Technical Report

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Introduction

This report outlines the methods and execution of the first assignment for the Semester3-ES class. The task is to understand bare-metal GPIO programming on the Nucleo F303RE microcontroller.

The requirements are the following:

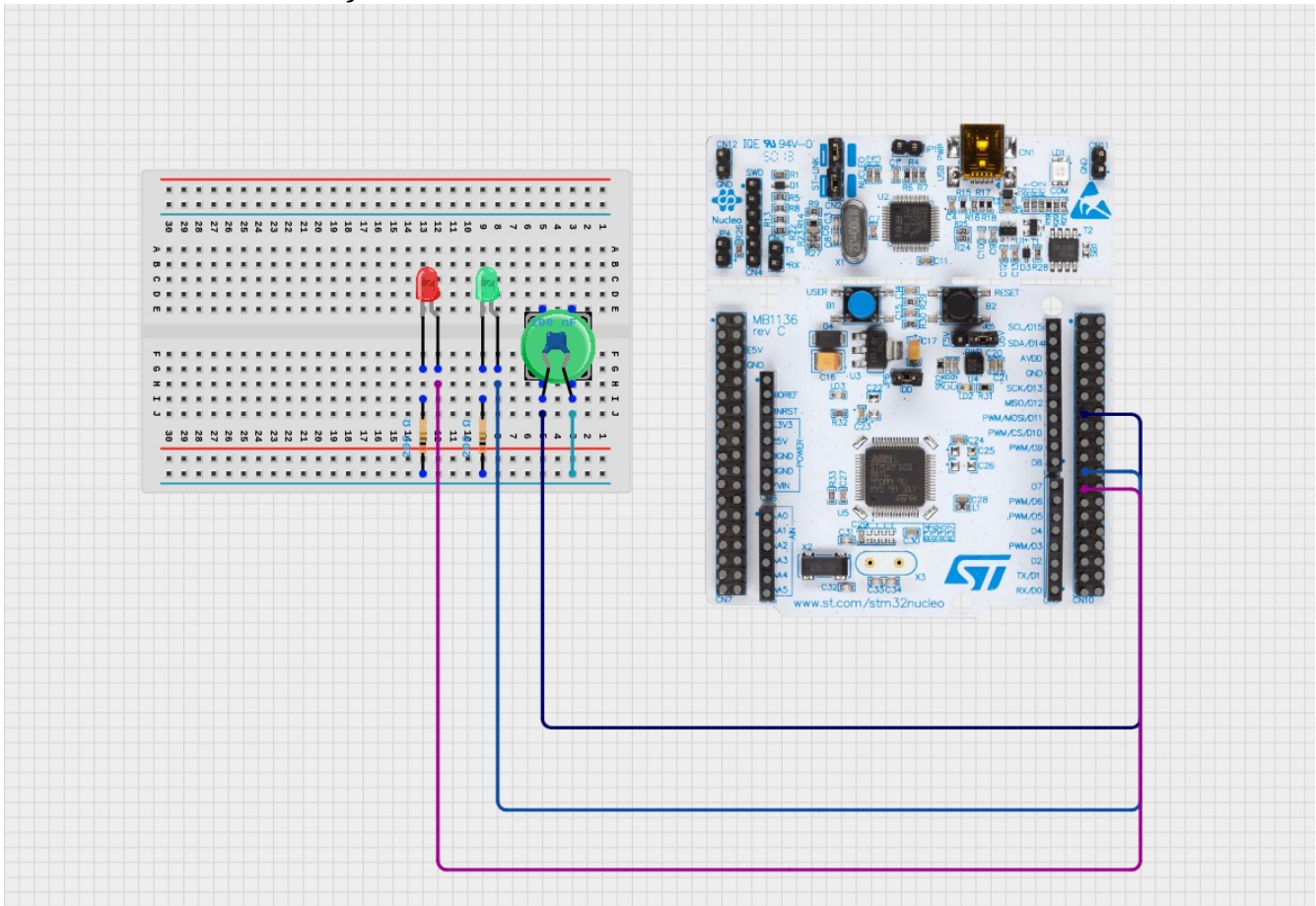
- Connect two buttons and two LEDs
- When pressing the first button, it must toggle its corresponding LED.
- When the second button is pressed, the other LED lights up and turns off after a short period of time.
- The last task is to create something fun. In this case I chose, when both buttons are pressed, the second(green) LED to light up in a pattern, spelling “Hello!” in morse code.

Build design

The circuit used for this experiment is the following:

- Red LED connected to PA8
- Green LED connected to PA9
- PushButton Connected to PA6 (debounced with a 100nf capacitor)

- Additionally, the built-in button on the microcontroller is used as the other button, internally it is connected on PC13.



Evidence

- When the USER button is pressed the light on PA8 toggles

```
LED on PA8 toggled!
LED on PA8 toggled!
```

- When the button on PA6 is pressed the LED on PA9 turns on for 3 seconds and then turns off

```
Turning PA9 on!
Turning PA9 off!
```

- When both buttons are pressed the green LED starts spelling “Hello” in morse code.

```
Running morse code!(.... . .-.. .-.. ---)
```

Conclusion

For a first assignment, it was quite pleasant. I learned how to operate the GPIO on the Nucleo and also I learned a lot of embedded system concepts and how things work and why they must be set up (like the clock for example).